



DEVELOPER HANDOVER DOCUMENT

Version 1.0 | Complete Feature Specification
Build from Scratch — Reference Implementation Provided

Project Name	AquaOps — Water Filter Business Management System
Document Type	Developer Handover / Feature Specification
Version	1.0 — Initial Release
Prepared For	Development Team
Tech Stack	HTML + CSS + JavaScript (Frontend) / Backend TBD by developer
Data Storage	localStorage (prototype) → migrate to real database
Status	Prototype Complete — Ready for Production Build

1. Project Overview

AquaOps is a complete business management system for a water filter shop. It handles customer lifecycle management, filter expiry tracking, WhatsApp-based notifications, installation scheduling, inventory tracking, employee management with individual logins, and customer feedback collection.

Core Business Problem Being Solved

Water filter businesses lose recurring revenue because they have no system to track when customer filters expire. Without reminders, customers forget to replace filters, and the shop loses replacement jobs. AquaOps fixes this with automated expiry tracking and one-click WhatsApp reminders.

1.1 Business Context

- Retail & Installation: Shop sells filters and sends technicians to install at customer sites
- Recurring Maintenance: Each filter has a lifespan (e.g. 6 months). When it expires, the customer needs a replacement job — this is the core recurring revenue
- Team: Manager, Field Technicians, and In-Shop Sales Staff — each with different access levels
- Inventory: Filter cartridges and spare parts must be tracked and restocked before they run out

1.2 User Roles

Role	Access Level	Primary Responsibilities
Manager	Full Access — all modules	Oversee everything: customers, expiry, scheduling, inventory, staff, reports, WhatsApp notifications
Technician	Limited — own jobs only	View assigned jobs, log parts used, mark jobs complete, view job addresses
Sales Staff	Limited — customer-facing	Make reminder calls, log daily sales, view customer records, send feedback requests

2. Login & Authentication System

2.1 How Login Works

The login flow is a two-step process designed for simplicity on mobile and tablets used in the shop:

1. Employee types their Employee ID (e.g. EMP-001) into the input field
2. System instantly shows a preview card with the employee's name and role (live lookup — no submit button needed)
3. Employee then enters their 4-digit PIN on a large keypad
4. If PIN matches: employee lands on their role-specific dashboard
5. If PIN is wrong: keypad shakes with red dots, resets automatically — employee tries again

2.2 Login Screen Features

- Employee ID input with live lookup — shows name + role preview before PIN entry
- Large on-screen PIN keypad (works on touchscreen tablets in the shop)
- 4 animated PIN dots — fill as digits are entered, shake red on wrong PIN
- Keyboard support — can type numbers and press Enter on a laptop
- "Change Employee ID" back link — returns to ID step without page reload
- Session timer visible in header — shows how long the employee has been logged in
- Every login and logout is recorded in an Activity Log with timestamp

2.3 Manager Controls for Accounts

- Add new employee account (name, phone, role, shift, area, 4-digit PIN)
- Reset any employee's PIN (manager enters new 4-digit PIN)
- Disable / Enable any employee account (disabled accounts cannot log in)
- View all employees with their last login time and job count this month
- View full activity log — who logged in, when, from what role

Developer Note — Authentication

The prototype stores PINs in plain text in `localStorage`. For production, implement proper authentication: hash PINs with `bcrypt`, use JWT tokens or sessions, and store all user data in a server-side database. The employee ID + PIN pattern should remain the same — only the backend needs to change.

3. Manager Dashboard — Full Feature Specification

The Manager Dashboard is the most feature-rich part of the system. It contains 8 sections accessible via a left sidebar navigation.

3.1 Main Dashboard (Overview Page)

Stat Cards (5 cards at top)

Card	What It Shows	Data Source
Total Customers	Count of all registered customers	customers table
Overdue	Customers whose filter has already expired	daysUntil(due) < 0
Due This Month	Customers whose filter expires within 30 days	daysUntil(due) <= 30
Jobs Done	Completed jobs in the current calendar month	jobs where status=Completed, month=current
Low Stock	Inventory items at or below their reorder level	stock <= reorder_level

Dashboard Widgets (4 panels below stat cards)

- **Urgent Panel** — Lists top 5 overdue + due-within-7-days customers sorted by urgency. Each row shows days remaining, customer name, filter model, and a 📱 WhatsApp button to notify immediately
- **Today's Installations** — Shows all jobs scheduled for today sorted by time. Shows time, customer name, job type, assigned technician, and status pill
- **Inventory Alerts** — Shows all items below reorder level with a "🔴 Out of Stock" or "🟡 Reorder" icon, current quantity, and a quick "+ Stock" button
- **Latest Feedback** — Shows the 3 most recent customer feedback entries with star rating and comment preview

3.2 All Customers Page

Customer Table Columns

Column	Description
Customer ID	Auto-generated (CUST-001, CUST-002, etc.)

Column	Description
Full Name	Customer full name
Phone	Phone number with country code (used for WhatsApp links)
Filter Model	Which filter was installed (e.g. Standard 5-Stage RO)
Install Date	Date of original installation
Due Date	Auto-calculated: Install Date + Lifespan months
Status	Color-coded pill: Active (green) / Due soon (yellow/orange) / Overdue (red)
Technician	Which technician did the installation

Actions on Each Customer Row

-  WhatsApp button — opens WhatsApp with pre-filled filter expiry reminder message
-  Feedback button — opens WhatsApp with pre-filled feedback request message
-  Delete button — removes customer from system (with confirmation)

Add Customer Form Fields

Field	Type	Notes
Full Name	Text input	Required
Phone	Text input	With country code for WhatsApp links (e.g. 923001234567)
Address	Text input	Full service address
Area / Zone	Text input	Used to assign technician by area
Filter Model	Dropdown	Standard 5-Stage RO, Ultra 7-Stage RO, Basic Sediment, Carbon Block, UV Sterilizer Combo, Whole House Filter
Lifespan (Months)	Number	e.g. 6 months — due date auto-calculated from this
Install Date	Date picker	Defaults to today
Technician	Dropdown	Populated from active Technician employees
Notes	Textarea	Optional special instructions

Search & Filter

- Live search bar filters customer table by name or phone number in real time (no page reload)



3.3 Expiry Alerts Page

This is the most critical page for generating recurring revenue. It shows all customers whose filter has expired or will expire within 30 days.

Top Stat Cards

- Overdue count — filters already expired
- This Week — expiring within 7 days
- This Month — expiring within 30 days

Customer Reminder Cards

Each customer is shown as a card with:

- Large number showing days remaining (red = overdue, orange = urgent, yellow = due soon)
- "OVERDUE" or "DAYS LEFT" label below the number
- Customer name, phone, area, filter model, due date, assigned technician
-  **WhatsApp button** — opens WhatsApp with appropriate message (Overdue Urgent template if expired, Filter Expiry template if upcoming)
-  **Book Job button** — pre-fills the Schedule Job form with this customer's details
-  **Feedback button** — sends feedback request via WhatsApp

"WhatsApp All Overdue" Button

- Single button at top of page opens WhatsApp for every overdue customer sequentially
- Uses 300ms delay between each to avoid browser blocking
- Logs each send to the WhatsApp Sent Log



3.4 WhatsApp Notifications Page

Message Templates (5 pre-built templates)

Template Name	When to Use	Key Content
Filter Expiry	Customer filter due in 1–30 days	Personalized greeting, days remaining, booking call-to-action, shop name

Template Name	When to Use	Key Content
Booking Confirmation	After scheduling a job	Customer name, date, time, technician name, instructions
Feedback Request	After completing a job	Thank you message, star rating options (★ to ★★★★★), reply instructions
Overdue Urgent	Filter already expired	Urgent warning, health risk message, call-to-action
Special Offer / Promo	Marketing campaigns	Promotional message with offer details

Sending Flow

6. Manager selects a message template (highlighted with blue border)
7. Manager selects one or multiple customers from checkbox list on the left
8. Preview panel on the right shows the exact message that will be sent to first selected customer
9. Click "Send" button — opens WhatsApp with pre-filled message for each customer
10. Each send is logged in the Sent Log table with customer name, template used, time, and date

Customer Selection Controls

- "Select All" button — checks all customers
- "Clear" button — unchecks all
- Individual checkboxes with name, phone, and days-remaining badge visible

Sent Log Table

- Shows last 20 sent messages
- Columns: Customer Name | Template | Time & Date | Status (always "Sent")

★ 3.5 Customer Feedback Page

Feedback Stats (4 cards)

- Average Rating (calculated from all feedback entries)
- Total Reviews count
- 5-Star count
- Complaints count

Feedback Cards

- Customer name and phone
- Star rating displayed as ★ characters
- Feedback type pill: Compliment (green) / Suggestion (blue) / Complaint (red)
- Customer comment text
- Date of feedback
- ➡️ Follow-up button — opens WhatsApp feedback request to same customer

Record Feedback Form

Manager can manually log feedback received over phone or in person:

Field	Type	Options
Customer Name	Text	Free text
Phone	Text	For follow-up WhatsApp
Rating	Dropdown	5 Star Excellent / 4 Star Good / 3 Star Average / 2 Star Poor / 1 Star Very Poor
Type	Dropdown	Compliment / Suggestion / Complaint
Comment	Textarea	What the customer said



3.6 Installation Schedule Page

Filter Tabs

- Today — jobs scheduled for today only
- Tomorrow — next day's jobs
- This Week — next 7 days
- All Upcoming — all future scheduled jobs

Timeline View (left panel)

- Vertical timeline with each job as a dot on a line
- Shows time (formatted as 10:00 AM), date, customer name, job type, technician, address
- Completed jobs shown with green dot, scheduled with blue dot

Jobs Table (right panel)

Column	Description
Time	Scheduled time formatted (e.g. 10:00 AM) with date below
Customer	Name with phone number below
Type	New Installation / Filter Replacement / Repair / Inspection
Technician	Assigned technician name
Status	Scheduled / In Progress / Completed / Cancelled
Actions	✓ Done button (marks complete), ➡ WhatsApp booking confirmation, ✕ Cancel

Schedule Job Form Fields

Field	Type	Notes
Customer Name	Text	Required
Phone	Text	With country code for WhatsApp
Address	Text	Service address
Job Type	Dropdown	New Installation / Filter Replacement / Repair / Inspection
Technician	Dropdown	Active technicians from employee list
Date	Date picker	Defaults to today
Time	Time picker	Defaults to 10:00
Labor (PKR)	Number	Labor charge in Pakistani Rupees

Field	Type	Notes
Parts (PKR)	Number	Parts cost
Notes	Textarea	Optional



3.7 Inventory Page

Inventory Table

Column	Description
Item Name + Supplier	Product name with supplier name below
Category	Filter Cartridge / Membrane / Housing / Fittings / Tubing / Tank / Tools / Other
Stock Level Bar	Visual progress bar showing stock as percentage of (reorder level × 3)
Current Stock	Number with unit (Pcs, Sets, Meters)
Reorder At	Threshold number — triggers alert when stock hits this
Status	OK (green) / Reorder (yellow) / Out of Stock (red)
Actions	+ Stock button (prompts for quantity, adds to stock) / × Delete

Add Inventory Item Form

Field	Type	Notes
Item Name	Text	Required
Category	Dropdown	Filter Cartridge, Membrane, Housing, Fittings, Tubing, Tank, Tools, Other
Current Stock	Number	Opening stock quantity
Reorder Level	Number	Alert triggers when stock reaches this number

Field	Type	Notes
Unit	Dropdown	Pcs / Sets / Meters
Supplier	Text	Supplier name for reference



3.8 Employees Page

Employee Table

Column	Description
Employee ID	Auto-generated (EMP-001, EMP-002, etc.)
Name	With avatar showing first letter in role color
Role	Manager (amber) / Technician (blue) / Sales Staff (grey)
Phone	Contact number
Shift	Morning / Evening / Night
Area	Zone assignment
Jobs/Month	Count of completed jobs by this technician in current month
Status	Active (green) / Inactive (grey)
Actions	PIN Reset / Disable or Enable account

4. Technician Dashboard — Feature Specification

When a Technician logs in, they see a simplified dashboard showing only what they need. Management features are locked and shown as greyed-out with a  icon.

4.1 Accessible Modules

Module	What Technician Can Do
My Jobs	See all jobs assigned to them. View customer name, address, job type, scheduled time. Mark job as complete after finishing on-site.
Parts Log	Record parts used on each job. This auto-deducts from inventory so stock is always accurate.
Job Locations	View customer addresses for today's assigned jobs. Helps with route planning.
Session History	Technician can see their own login/logout history with times and dates.

4.2 Locked Modules (visible but inaccessible)

- Reports — Manager only
- Employees — Manager only
- Settings — Manager only
- Inventory (full view) — Manager only. Technician can only log parts used on a job.

Developer Note — Technician Dashboard

The Technician dashboard in the prototype shows the access structure. Full implementation should include: their own jobs list with complete/incomplete status, a form to submit parts used (which deducts inventory), and a map link to the job address.

5. Sales Staff Dashboard — Feature Specification

Sales Staff work in the shop. They handle reminder calls, log in-shop sales, and can view customer records. They cannot access financial reports, employee management, or full inventory.

5.1 Accessible Modules

Module	What Sales Staff Can Do
Reminders	See today's customers who need a call for filter replacement. Make calls, log results (booked / no answer / refused), set follow-up dates.
Daily Sales	Log in-shop sales: item sold, quantity, unit price, payment method, customer name.
Customers	View customer records (read-only). Cannot add, edit, or delete customers.
My Jobs	View their own session history (login/logout times).

6. WhatsApp Message Templates — Full Content

All messages open WhatsApp via the wa.me link format:

https://wa.me/{phone_number}?text={url_encoded_message}

Template 1: Filter Expiry Reminder

Assalamualaikum **{customer_name}** 🖐️

Your water filter is due for replacement in **{days} days**.

🔧 Keeping your filter fresh ensures clean, safe drinking water for your family.

 ***Book your replacement now*** — our technician will visit at your convenience.

Reply ***YES*** to confirm booking or call us anytime.

AquaOps Water Solutions 💧

Template 2: Booking Confirmation

Assalamualaikum **{customer_name}** 🖐️

✅ ***Your appointment is confirmed!***

 Date: **{date}***

🕒 Time: **{time}***

👷 Technician: **{technician_name}***

Please ensure someone is home at the scheduled time.

For any changes, reply to this message.

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Template 3: Feedback Request

Assalamualaikum **{customer_name}** 🖐️

Thank you for choosing *AquaOps Water Solutions*! 💧

We hope our service met your expectations. Could you spare 30 seconds to share your experience?

Please reply with:

★★★★★ — Excellent

★★★★ — Good

★★★ — Average

★★ — Poor

Your feedback helps us serve you better. Thank you!

Template 4: Overdue Urgent

🚨 ***Urgent Notice — {customer_name}***

Your water filter has **expired** and needs immediate replacement.

⚠️ An expired filter may no longer remove harmful contaminants from your water.

☎️ Please call us **today** or reply to this message to schedule an urgent visit.

We are here to help! 💧

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Template 5: Special Offer / Promo

Assalamualaikum ***{customer_name}*** 🙌

🎁 ***Special Offer — This Month Only!***

Get your filter replacement done and receive **FREE servicing** on your next visit.

💧 Clean water is priceless — take advantage of this offer today!

Reply *OFFER* to claim or call us to book.

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7. Database / Data Models

The prototype uses localStorage. In production, implement these as database tables (MySQL, PostgreSQL, or Supabase/Firebase for quick deployment).

7.1 Customers Table

Field	Type	Description
id	String / UUID	Auto-generated (CUST-001 or UUID)
name	String	Customer full name
phone	String	With country code, no spaces or dashes (for wa.me links)
address	String	Full service address
area	String	Zone/area for technician assignment
model	String	Filter model installed
life	Integer	Filter lifespan in months
install	Date	Installation date
due	Date	Calculated: install + life months. Auto-recalculate on update.
tech	String	Assigned technician name (FK to employees)
notes	Text	Optional special instructions
created_at	DateTime	Record creation timestamp

7.2 Jobs Table

Field	Type	Description
id	String / UUID	Auto-generated (JOB-001 or UUID)
cust	String	Customer name (FK to customers)
phone	String	Customer phone
addr	String	Service address
type	Enum	New Installation / Filter Replacement / Repair / Inspection

Field	Type	Description
tech	String	Assigned technician (FK to employees)
date	Date	Scheduled date
time	String	Scheduled time (HH:MM)
status	Enum	Scheduled / In Progress / Completed / Cancelled
labor	Decimal	Labor charge in PKR
parts	Decimal	Parts cost in PKR
payment	Enum	Unpaid / Paid / Pending Payment
completedDate	Date	Filled when status becomes Completed
notes	Text	Optional notes

7.3 Inventory Table

Field	Type	Description
id	String / UUID	Auto-generated
name	String	Item name
cat	String	Category
unit	String	Pcs / Sets / Meters
stock	Decimal	Current stock quantity
reorder	Decimal	Reorder threshold
supplier	String	Supplier name
last_restocked	Date	Last time stock was added

7.4 Employees Table

Field	Type	Description
id	String / UUID	Auto-generated (EMP-001 or UUID)

Field	Type	Description
name	String	Full name
role	Enum	Manager / Technician / Sales Staff
phone	String	Contact number
shift	Enum	Morning / Evening / Night
area	String	Assigned zone/area
pin	String	4-digit PIN — store as bcrypt hash in production
active	Boolean	Whether account can log in
created_at	DateTime	Account creation timestamp

7.5 Feedback Table

Field	Type	Description
id	String / UUID	Auto-generated
name	String	Customer name
phone	String	Customer phone for follow-up
rating	Integer	1–5 star rating
type	Enum	Compliment / Suggestion / Complaint
comment	Text	Customer comment
date	Date	Date feedback was recorded

7.6 WhatsApp Log Table

Field	Type	Description
id	Auto	Auto-generated
name	String	Customer name message was sent to
tpl	String	Template used (expiry / booking / feedback / overdue / promo)

Field	Type	Description
time	String	Time string
date	String	Date string
sent_by	String	Employee ID who sent it

8. What Is Built vs. What Still Needs Building

8.1 Already Built (Prototype — HTML Files)

Two Complete HTML Files Delivered

File 1: step1-login.html — Complete login screen with PIN keypad, role preview, activity logging
File 2: step2-manager-dashboard.html — Complete Manager Dashboard with all 8 modules

- **Login Screen:** Employee ID lookup, PIN keypad, role preview, session logging, manager employee management
- **Manager Dashboard:** Dashboard overview, All Customers, Expiry Alerts, WhatsApp Notifications (5 templates), Feedback, Install Schedule, Inventory, Employees
- **Demo Data:** Seeded with realistic sample customers, jobs, inventory, and feedback for testing
- **WhatsApp Integration:** All 5 templates fully written and wired to wa.me links

8.2 Still To Build

Feature	Priority	Notes
Technician Dashboard	HIGH	My Jobs list, Parts logging form, job completion flow
Sales Staff Dashboard	HIGH	Reminder calls list with result logging, daily sales form
Real Backend / Database	HIGH	Replace localStorage with server + database (Supabase recommended for speed)
PIN Hashing (Security)	HIGH	Hash PINs with bcrypt before storing
Session Management	HIGH	JWT tokens or server sessions instead of in-memory state
Shift Handoff Module	MEDIUM	Generates WhatsApp-ready handoff report for end of shift
Daily Sales Tracking	MEDIUM	Sales log page for in-shop sales with daily totals
Revenue Reports	MEDIUM	Monthly revenue, jobs per technician, top-performing areas
Parts Deduction on Job Completion	MEDIUM	When technician marks job done + enters parts, auto-deducts inventory
Push Notifications	LOW	Browser or app push notifications for overdue alerts (alternative to WhatsApp)

Feature	Priority	Notes
Mobile App (PWA)	LOW	Wrap as Progressive Web App for home screen install on phones/tablets
Google Sheets Sync	LOW	Export data to Google Sheets for owner who prefers spreadsheets

9. Technology Stack Recommendations

Option A — Fastest to Deploy (Recommended)

Layer	Technology	Why
Frontend	HTML + CSS + Vanilla JS (already built)	No framework needed — prototype is clean vanilla JS, easy to extend
Backend	Node.js + Express	Lightweight, fast to set up, same language as frontend
Database	Supabase (PostgreSQL)	Free tier available, built-in REST API, real-time updates, no DevOps needed
Authentication	Supabase Auth + custom PIN layer	Handle sessions with Supabase, add PIN check on top
Hosting	Netlify (frontend) + Supabase (backend+DB)	Both have free tiers, deploy in hours
WhatsApp	wa.me links (already implemented)	No API needed — free, works instantly

Option B — Full Control

Layer	Technology	Why
Frontend	React (convert current HTML)	Better state management as app grows
Backend	Node.js + Express	REST API
Database	PostgreSQL (self-hosted or AWS RDS)	Full control
Auth	JWT + bcrypt	Industry standard
Hosting	VPS (DigitalOcean / Hetzner)	Full control, predictable cost

Recommendation for This Business

Go with Option A. The business owner needs this running quickly without a large IT budget. Supabase gives a real database, authentication, and REST API with a free tier that comfortably handles a water

filter business with hundreds of customers. The existing HTML prototype can be connected to Supabase in 1-2 days of development work.

10. Development Notes & Important Details

10.1 WhatsApp Phone Number Format

All phone numbers must be stored WITHOUT spaces, dashes, or + symbols. Format: country code + number.

- Pakistan: 92 + 10 digits — e.g. 923001234567
- UAE: 971 + 9 digits — e.g. 971501234567
- The wa.me link format: <https://wa.me/923001234567?text=message>

10.2 Due Date Calculation

Due date = Install Date + Lifespan (months). Recalculate if either changes.

- Use EDATE equivalent: add lifespan months to install date month
- When a replacement is done, update the install date to today and recalculate due date
- Filter lifespan varies by model — allow custom lifespan per customer, not fixed by model

10.3 Status Thresholds

Days Until Expiry	Status	Color
< 0 (already expired)	Overdue	Red (#ef4444)
0 to 7 days	Urgent	Orange (#f97316)
8 to 14 days	Due Soon	Yellow (#f59e0b)
15 to 30 days	Upcoming	Blue (#3b82f6)
> 30 days	Active	Green (#10b981)

10.4 Security Requirements for Production

- Hash all PINs with bcrypt (salt rounds: 10) before storing
- Use HTTPS everywhere — do not deploy over plain HTTP
- Implement rate limiting on login endpoint (max 5 attempts, then 15-minute lockout)
- Use JWT tokens with expiry (8-hour sessions for shop staff)
- Validate all inputs server-side — never trust frontend-only validation
- Keep WhatsApp message sending client-side (wa.me links) — no API keys needed

10.5 Key UI/UX Decisions (Keep These)

- PIN keypad must be large — staff use it on tablets with their fingers
- Status colors must match throughout — same color = same meaning everywhere
- WhatsApp buttons open in new tab — do not navigate away from the dashboard
- Every destructive action (delete customer, cancel job) requires confirmation dialog
- Employee ID lookup is live (no submit button) — feels faster, more professional
- Session timer visible in header — manager can see how long each session has been active

AquaOps — Developer Handover Document

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